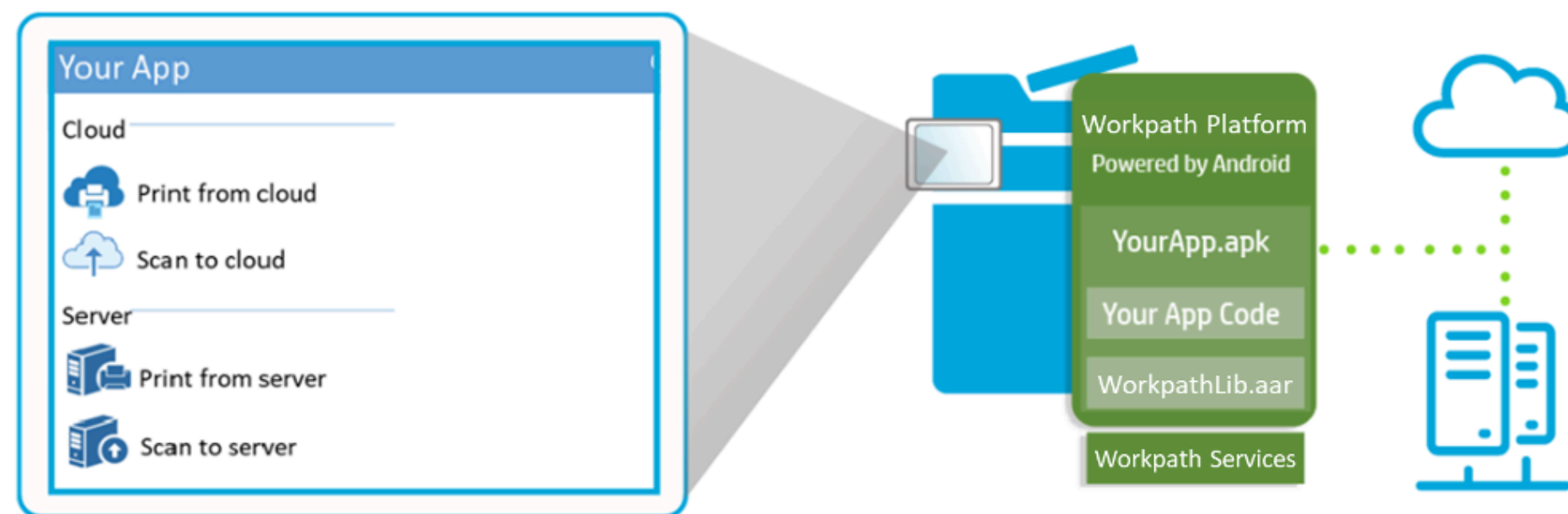


How It Works



The Workpath Platform

The Workpath Platform lets you install your app on an HP printer, where users will launch and use your app right from the device's control panel. Your app will use the Workpath library to utilize the scanning, printing, and other functions of the device.

You will develop your app using many of the same tools and technologies you already use to develop Android mobile apps. The device even supports remote debugging using the Workpath (Link) Debug Bridge (a version of ADB).

This page covers the following topics:

- SDK Installation and Setup
- Enabling the Workpath (Link) Platform
- Enabling the Workpath (Link) Debug Bridge (LDB)
- Installing and Debugging Your App
- UI Switching
- UI Style Guide: Tailoring Your App for HP Devices

SDK Installation and Setup

Developing Workpath applications requires installation of the following prerequisites:

- **SDK**
 - MinSdkVersion API 19
- **IDE** (Integrated Development Environment):
 - Android Studio 2.0.0 or above (Refer to <http://developer.android.com/sdk/index.html#Requirements> to set up the development environment)

How It Works

Compatible Devices

Platform Versions

Managing Client Credentials

Remote Configuration

Lighthouse Apps

Xamarin Binding

Enroll and Publish

Command Center & Test Flight

HP Proximity Card Readers

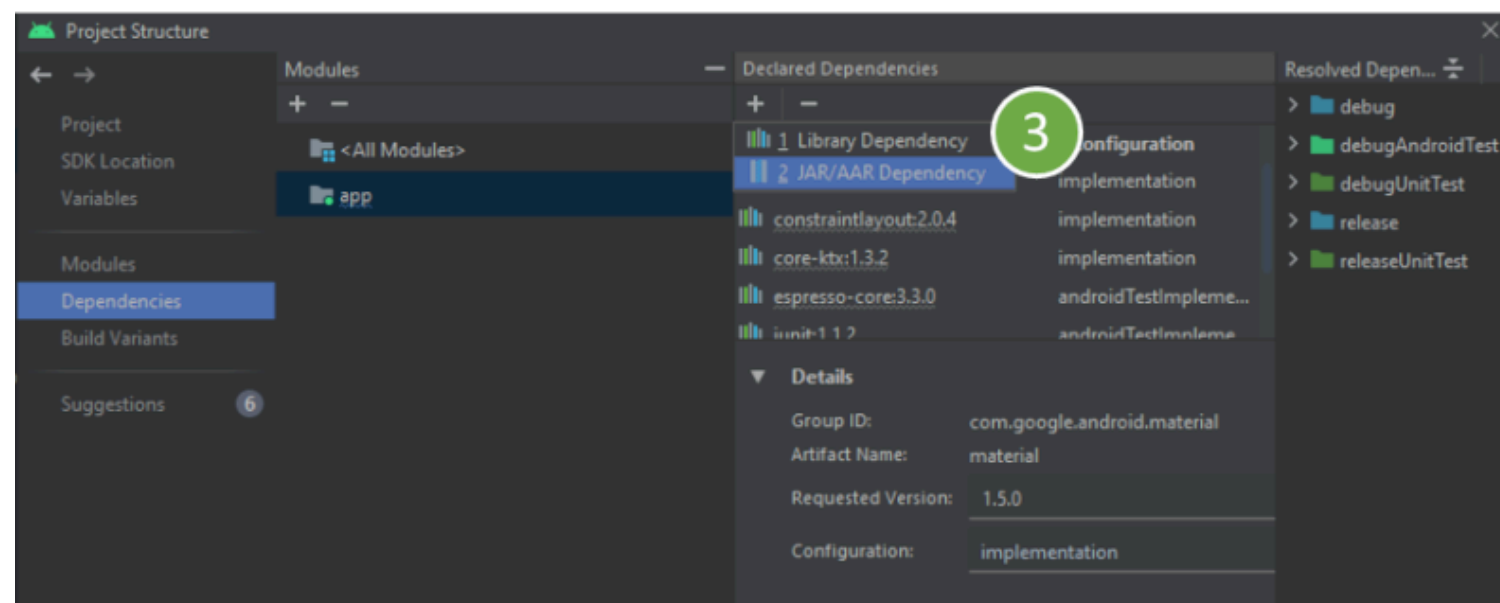
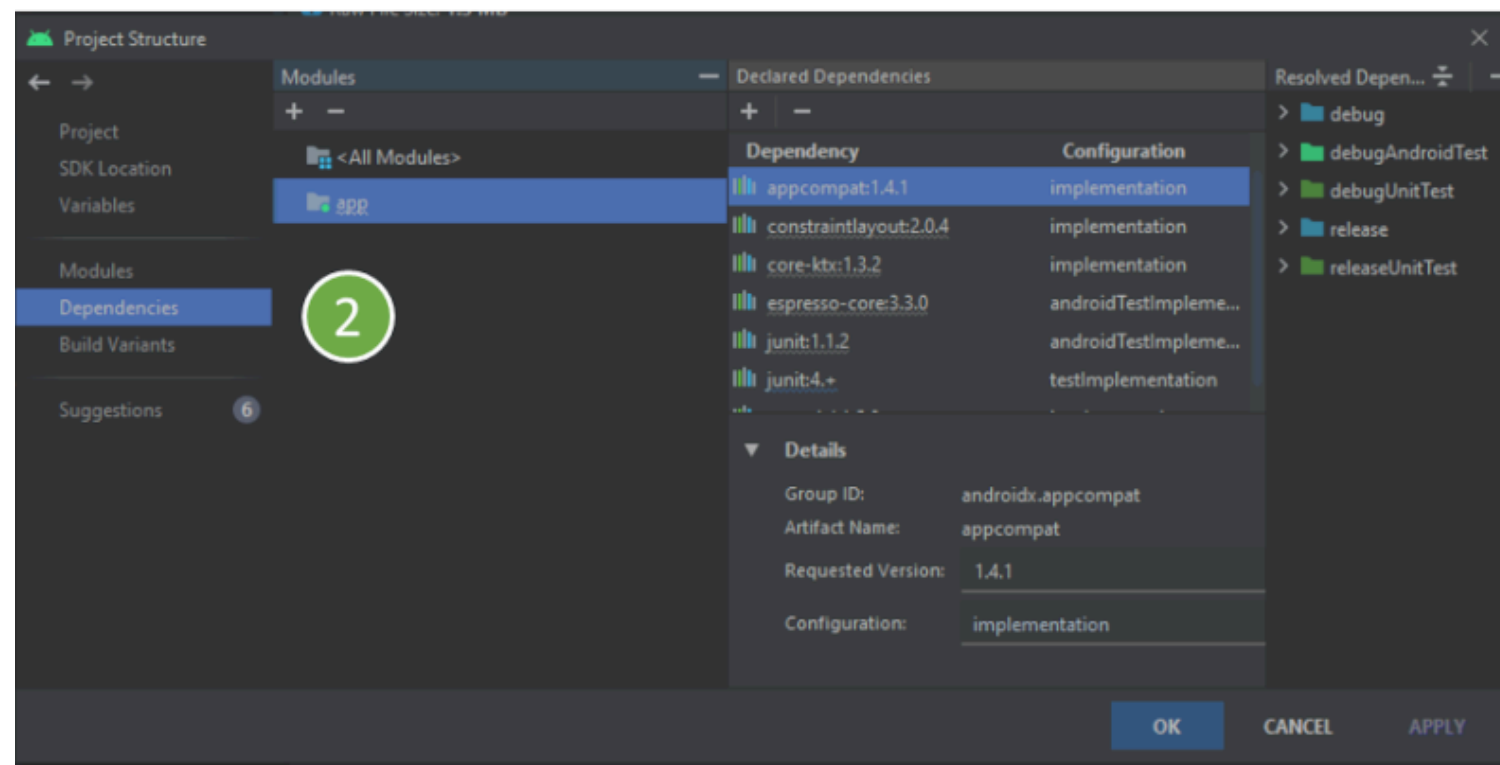
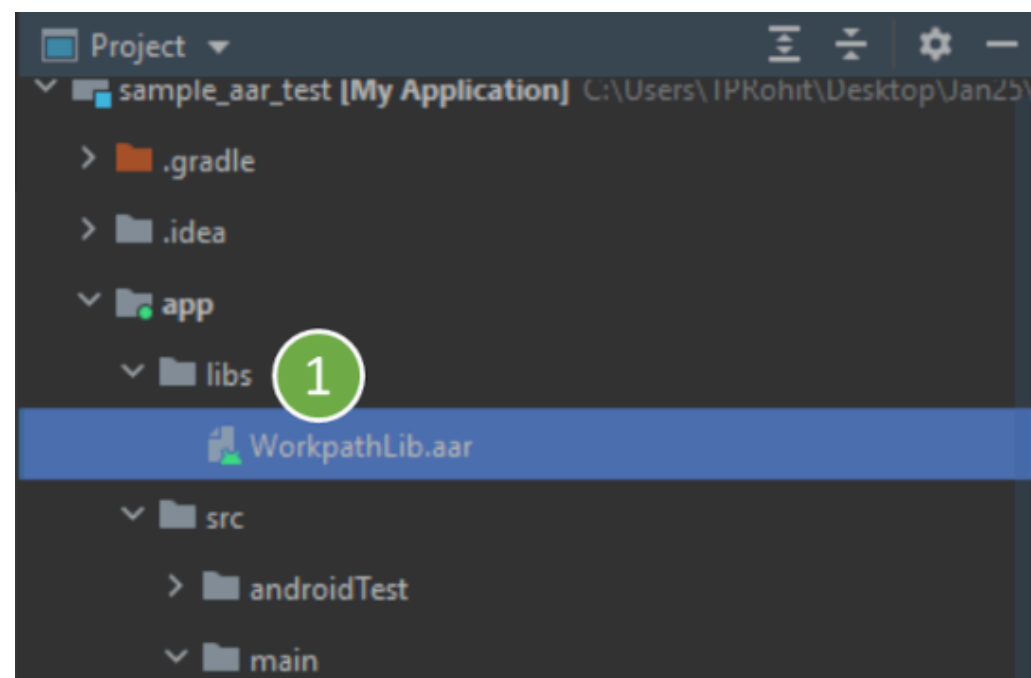
Self-Distribution

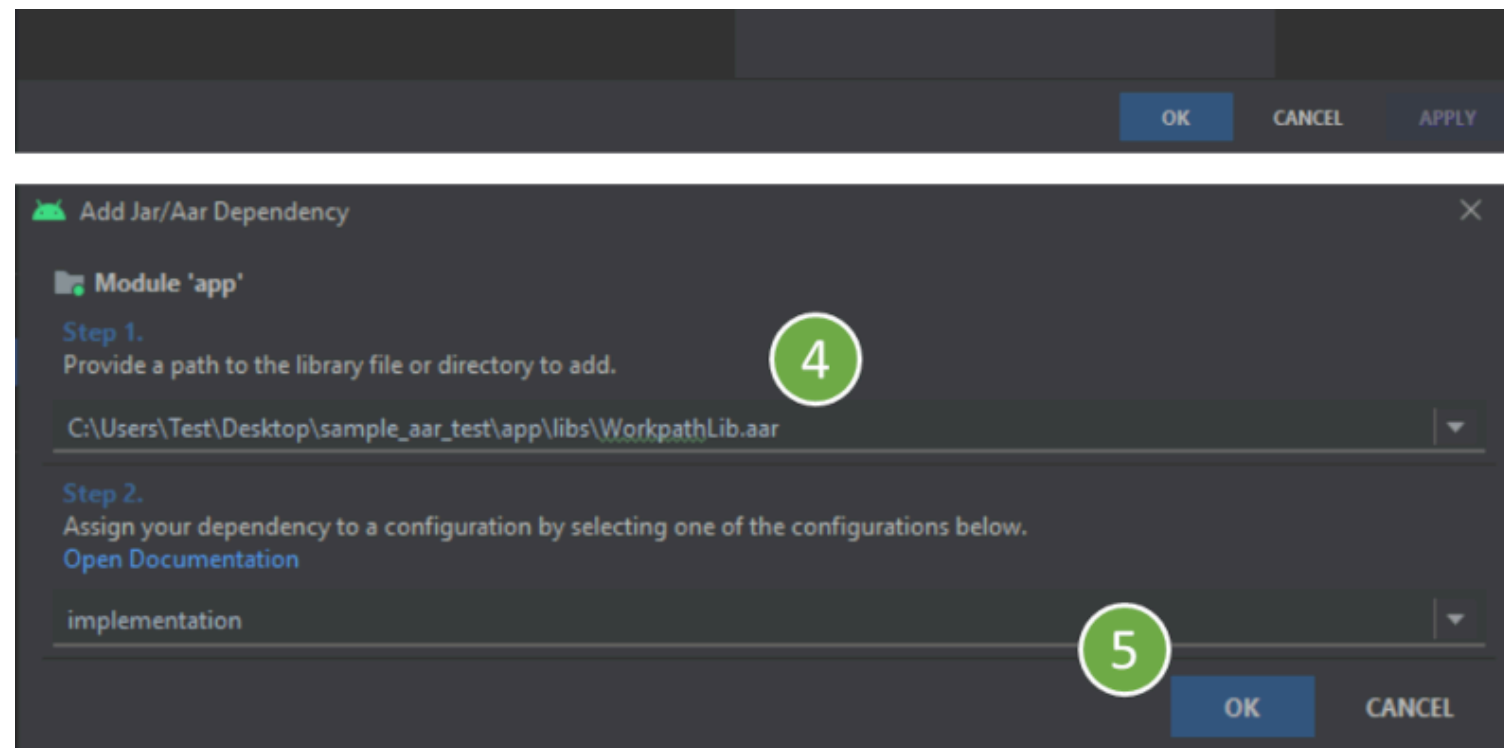
- **JDK** (Java Development Kit):
 - JDK 1.8 or higher

To use the Workpath API, you must import the the Workpath library (the WorkpathLib.aar located in Libraries folder of the SDK) and compile the project.

After downloading the SDK from the Downloads page and unzipping the SDK to your hard drive:

1. Select 'project' type > paste the WorkpathLib.aar file in the libs folder of your project
2. Right click on WorkpathLib.aar, click on Copy path > Copy absolute path, then Select File > Project structure
3. Select Declared Dependencies > click on '+' icon > 'JAR/AAR dependency'
4. Paste the path of the AAR copied earlier, set configuration as 'implementation'
5. Click OK, APPLY, OK, and wait for Gradle sync to complete





Enabling the Workpath (Link) Platform

The Workpath Platform is not enabled by default. To enable it, follow these steps:

- A. Check the list of compatible devices to be sure your device supports the Workpath Platform
- B. Upgrade the device to the latest firmware
- C. Enable the Workpath Platform using the device's EWS as follows:

1. Sign in
2. Navigate to the Security Tab
3. Navigate to the General Security page
4. Click on the Enable button in the Workpath (Link) section of the page
5. Click Restart at the bottom of the displayed page



HP Color LaserJet Flow E87660

HP Color LaserJet Flow E87660

Search by Keyword

Welcome
Sign In

Information

General

Copy/Print

Scan/Digital Send

Fax

Supplies

Troubleshooting

Security

HP Web Services

General Security

Account Policy

Access Control

Protect Stored Data

Manage Remote Apps

Certificate Management

Email Domain Restriction

Web Service Security

Self Test

General Security

Help

Enable host USB plug and play

Note: If not enabled, control panel apps that require host USB plug and play, such as Scan to USB Drive, will be automatically disabled.

HP JetAdvantage Link Platform

The HP JetAdvantage Link platform is currently disabled.

Enable

Printing

☐ Cancel print jobs after unattended error

Use this feature to prevent unauthorized users from printing jobs after clearing an error. All jobs are deleted from the print queue after the inactivity timeout period.

Status of Security Settings

Settings	Status
Device Password	Not configured
Service Access Code	Not configured
Remote Configuration Password	Not configured

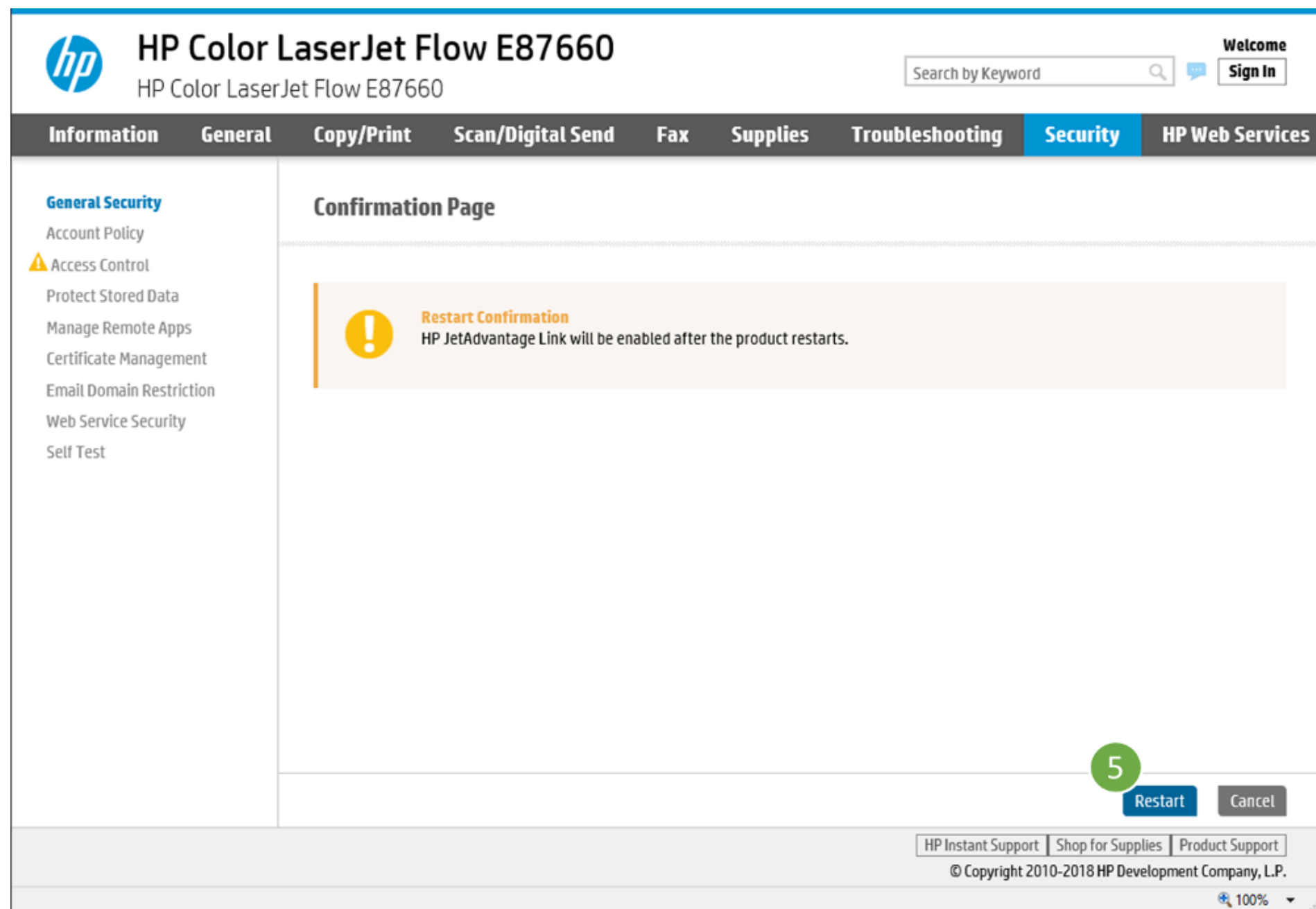
Apply

Cancel

HP Instant Support | Shop for Supplies | Product Support

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100%



Enabling the Workpath (Link) Debug Bridge (LDB)

The Workpath (Link) Debug Bridge (LDB) is a version of ADB. It allows you to install and debug your unsigned app on a device over the network from within Android Studio:

adb connect <device_ip_address>:5555

Warning: As a security measure, when enabling/disabling LDB, any currently installed apps will be automatically removed.

To Enable LDB, follow these steps:

- A. Create a personal LDB Service Key on the LDB Service Key page
- B. Connect your device to a network with internet access
- C. If required by your networking environment, use the EWS to configure the device to use your web proxy (See instructions here)
- D. LDB can only be enabled from the device's control panel. Enable LDB as follows:

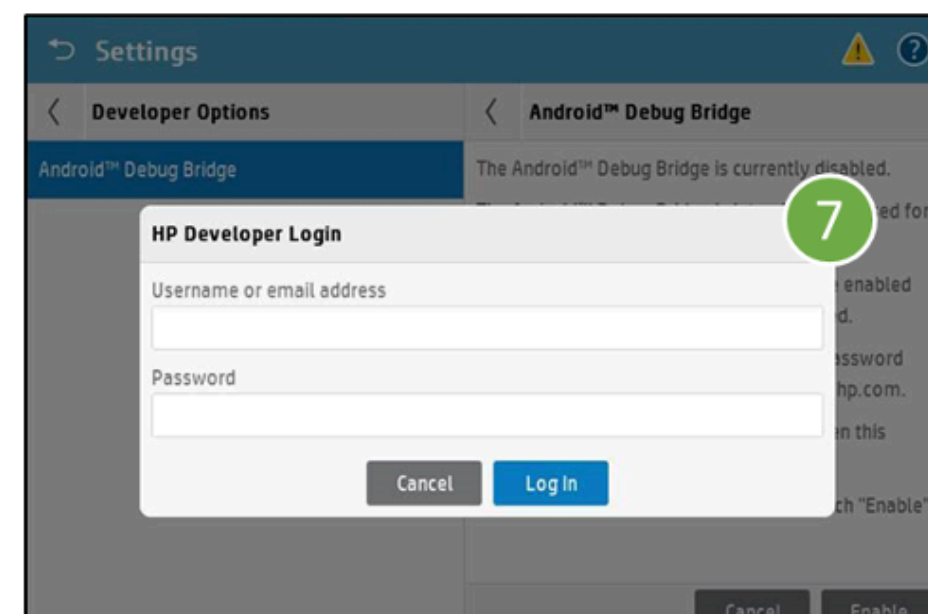
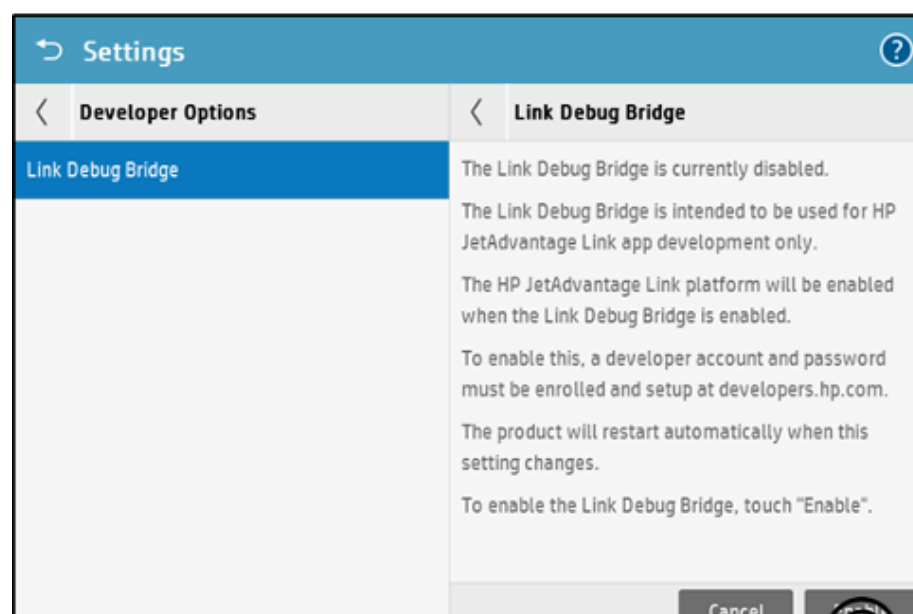
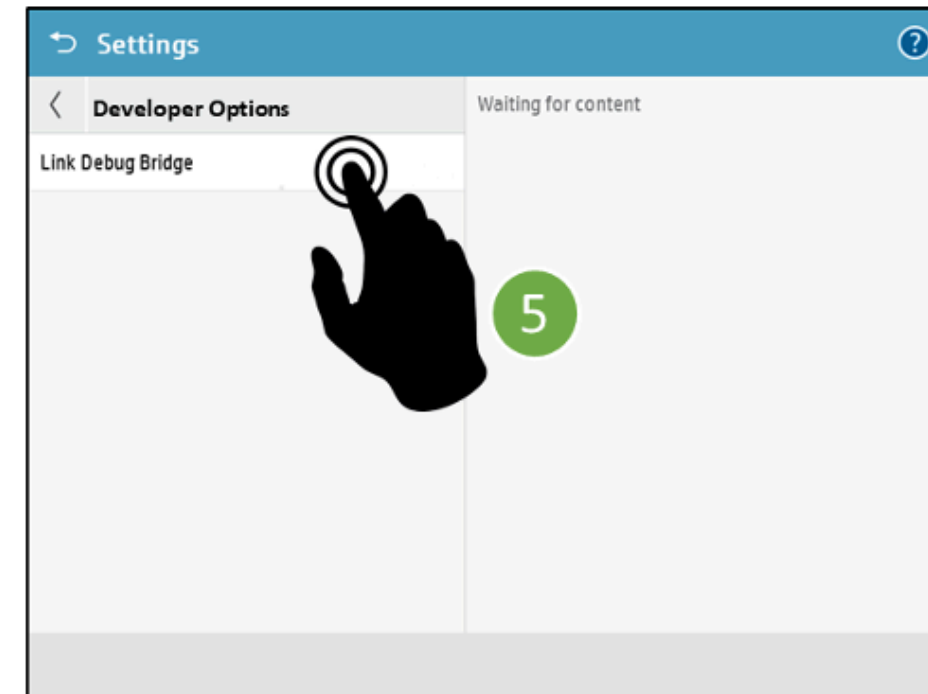
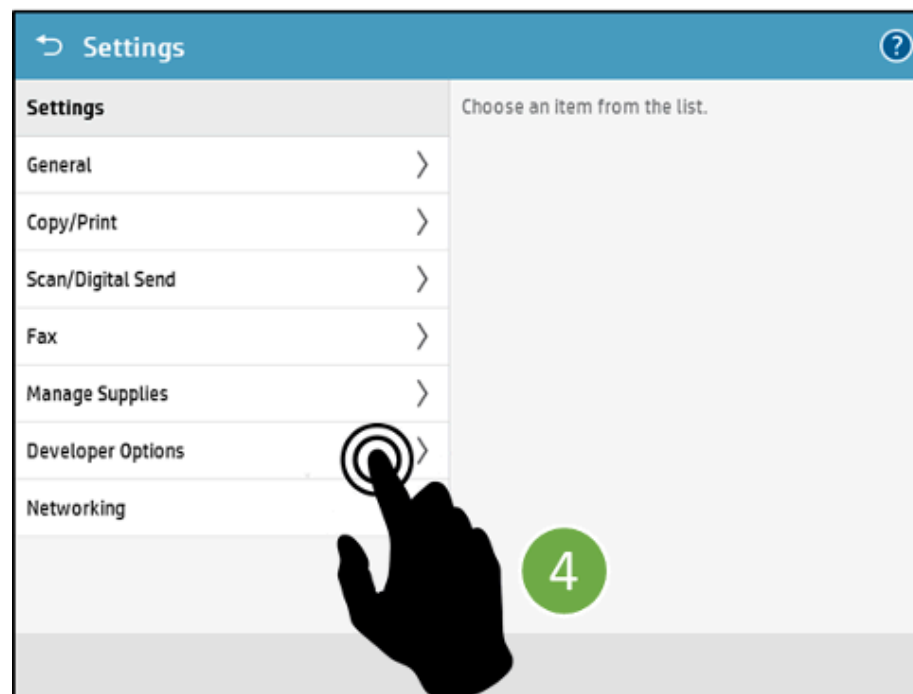
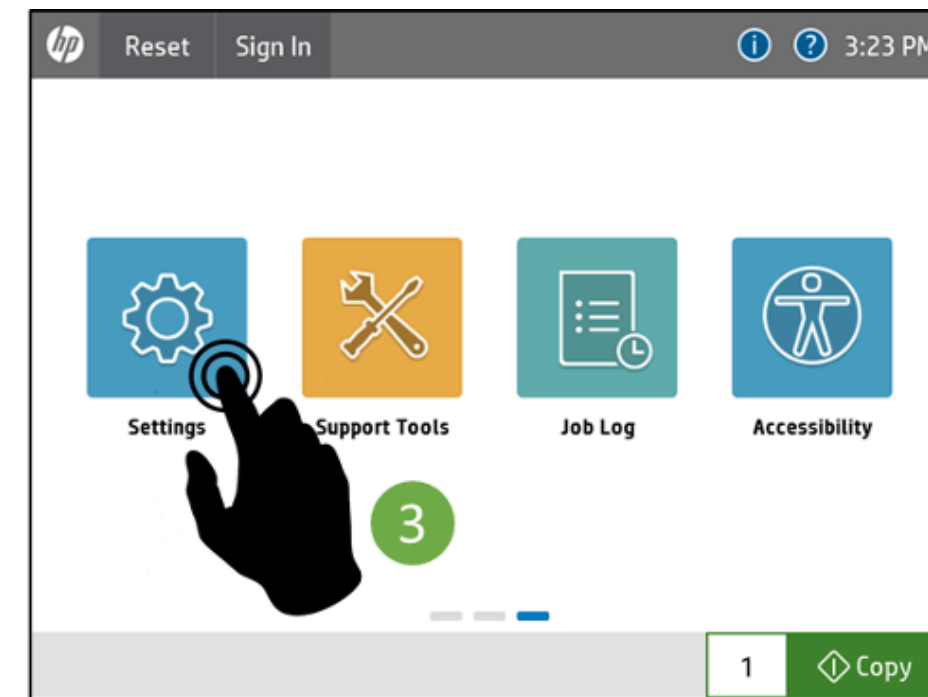
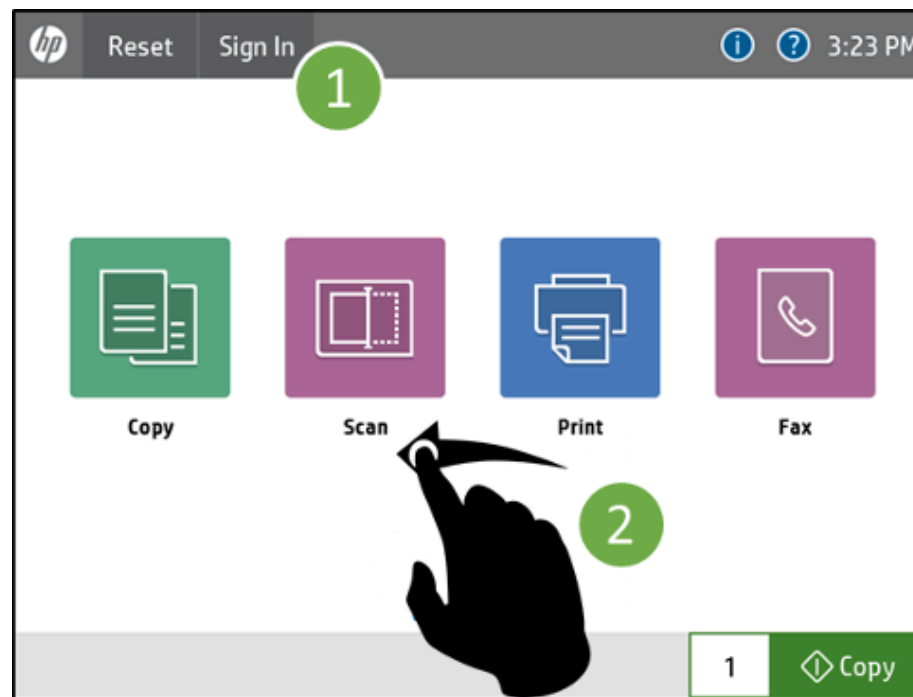
1. Sign into the device as an administrator
2. Swipe left until you see the Settings app

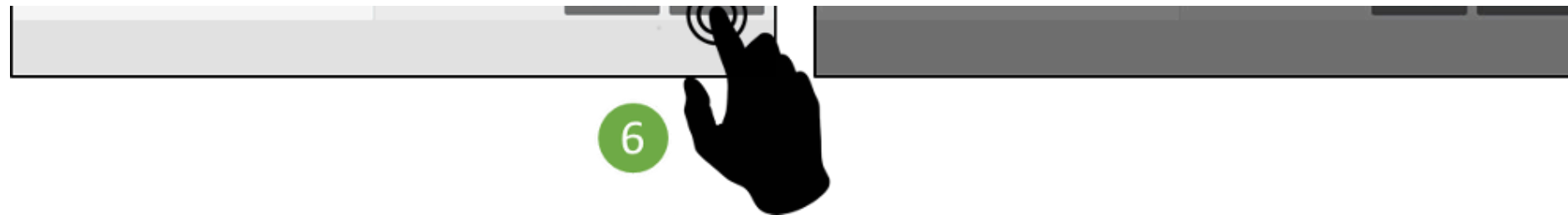
3. Click on the Settings app
4. Click on Developer Options
5. Click on Workpath (Link) Debug Bridge
6. Click on Enable
7. When prompted, enter your developers.hp.com account name (Username) and your LDB Service Key (Password)

Note: If the Workpath platform is not already enabled, it will be automatically enabled when LDB is enabled. If LDB is later disabled, the Workpath platform will remain enabled.

Note: As a security measure, the Workpath (Link) Debug Bridge menu cannot be accessed/enabled from within the Embedded Web Server.

Note: For your convenience during development, the Android Settings App is available in LDB mode. However, any changes made in the Android Settings App will not actually change the device's settings.





Installing and Debugging Your App

There are two ways to install your unsigned app on a device:

- Using the HPK tool
- Using LDB and Enhanced Debug Experience (EDX) Mode

Using the HPKTool

Any app intended for use on any HP device must be specially packaged (HPK file), submitted to HP for verification and validation (VAV), and finally signed by HP before it can be installed on a device (without LDB enabled). The SDK contains an HPKTool to make it easy to package your app in an HPK. A HPK file includes your signed APK and provides app-related information not present in a standard APK file. A HPK file intended for VAV and publishing must include the signed/release version of your APK. (**Note:** The APK must be signed using a V2 or V3 Signature.)

When used with a device where LDB is enabled, the HPK tool will allow you to install your unsigned HPK so you can test out the process before submitting your HPK to HP for VAV. Once installed, a button will be created on the device's control panel. (**Note:** You may need to press reset on the control panel before the button will appear.)

Once installed, you can launch your app from the front panel or from the adb console.

Using LDB and Enhanced Debug Experience (EDX) Mode

You can install your APK using LDB:

```
adb connect <device_ip_address>:5555
```

```
adb install <path_to_apk>
```

Note: Due to optimizations made by Android Studio for Instant Run, you cannot simply copy a debug APK out of Android Studio and install it. You must use either an APK generated from the command line or one specifically built using the “Build APK(s)” command from Android Studio’s “Build” menu. Release APKs will always work.

In normal mode, the Workpath Platform does not expose its own launcher. Instead, HP devices switch to the Workpath Platform when an app is launched and switch back when the app is closed. So when installing over LDB, you will not be able to launch your app directly from the HP home screen.

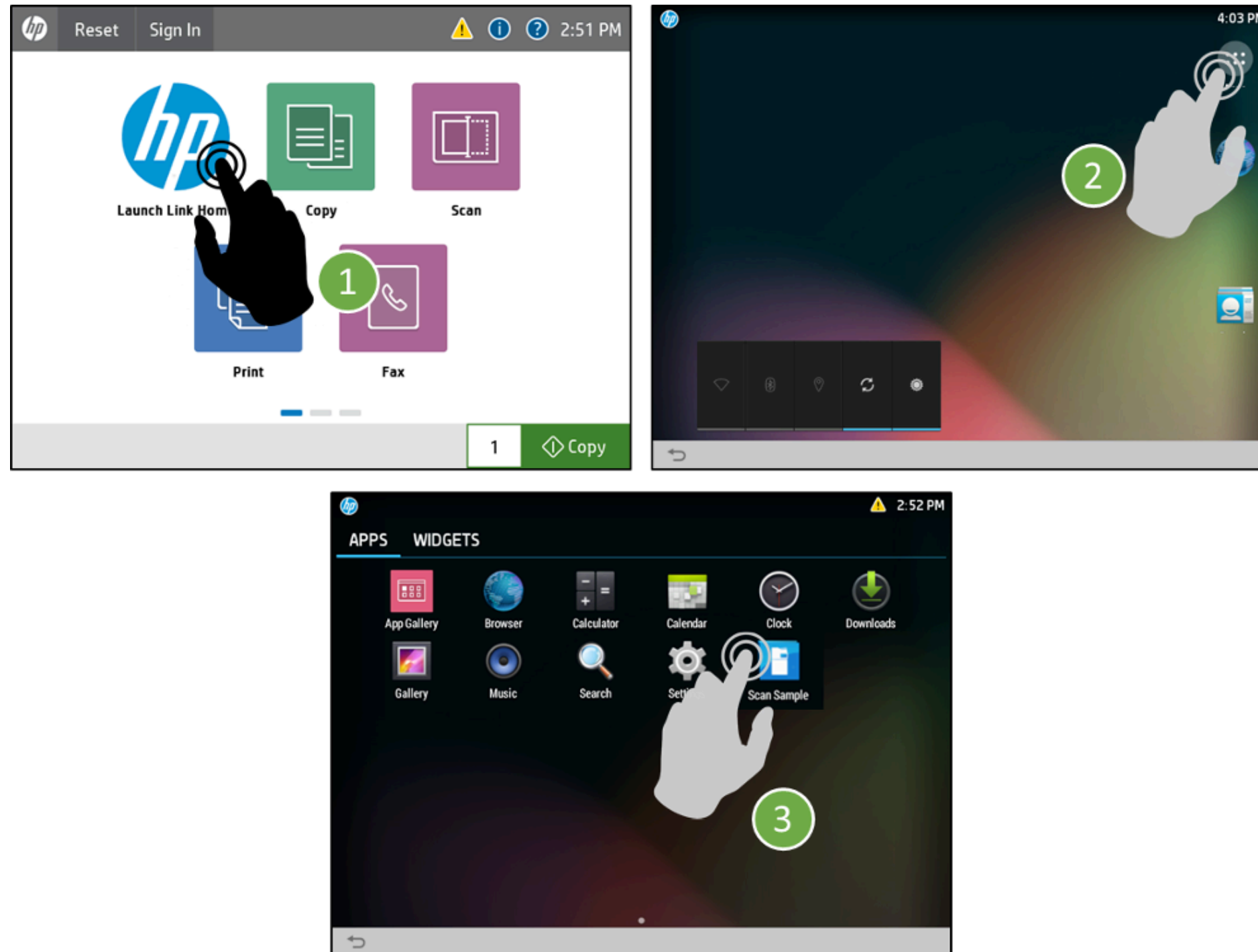
For your convenience, we have provided an Enhanced Debug Experience (EDX) mode that enables you to access the Workpath Platform's launcher just for app development. To turn on EDX mode, run the following command:

```
adb shell set_edx -v true
```

Note: With early firmware versions (prior to FS4.9), you will need to use one of the following variations of the adb command:

- `adb shell /system/bin/set_edx -v true`
- `adb shell sh /system/bin/set_edx -v true`

When EDX mode is enabled, a new "Launch Workpah (Link) Home" button will appear on the control panel to allow you to access the Workpath Platform launcher and from there launch your app:



You will also be able to load and debug your app using Android Studio's Instant Run functionality.

If you are debugging/testing with long breaks and need to disable the automatic UI timeout, you can establish a wake lock by running the following command:

```
adb shell set_edx -awake true
```

To disable EDX mode, run the following command:

```
adb shell set_edx -v false
```

Note: Disabling EDX mode will automatically release the wake lock.

To see all available options, run:

adb shell set_edx -help

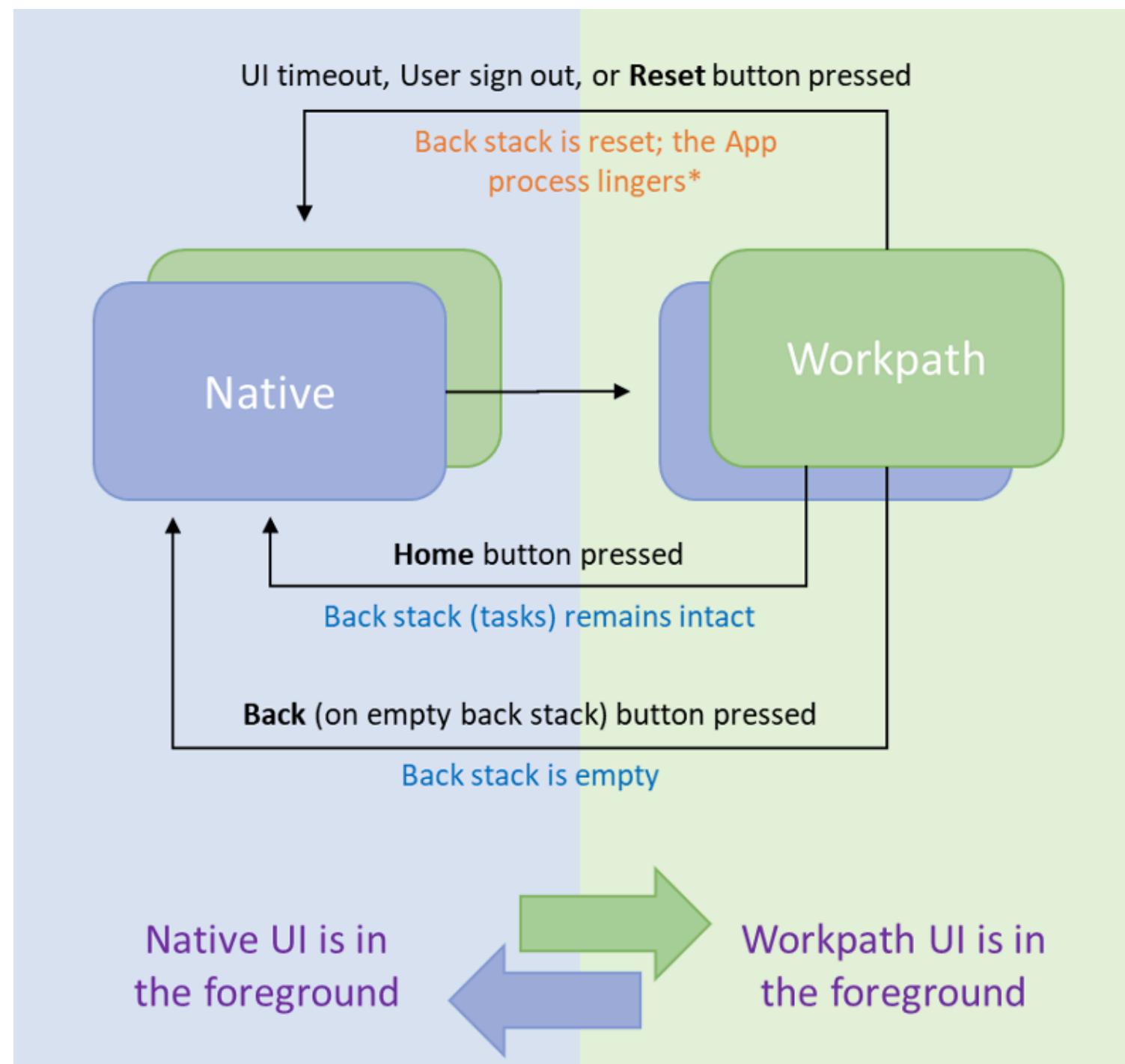
UI Switching

A Workpath app is represented as a button on the built-in (native) home screen. When that button is pressed, the device switches to the Workpath platform and launches the app. At this point, the app has the same UI abilities as android apps running on other platforms.

When the app is closed (by navigating back or by pressing home button), the device switches back to its native home screen. By design, apps are not expected to clear user session data on these transitions. This allows users to exit and reenter the app without losing their place.

The app can also be automatically closed by a reset condition (UI timeout, forced user sign-out, etc.) all of which also end the current user session. By design, apps are expected to clear user session data on these transitions.

It is important for you to understand these transitions, so you can implement your app appropriately. Please refer to the following diagram:



*A quick successive launch after this condition may reuse the existing lingering process. In such a case, the application will see all the data from the previous session which was attached to the process (e.g. singleton, any non terminated threads or services). App developers must detect this condition and perform necessary re-initialization logic.

UI Style Guide: Tailoring Your App for HP Devices

HP device control panel sizes differ from panel sizes commonly found on mobile devices, so you may need to adjust your screen layouts to look their best on these panel sizes.

For example, there is a specific recommended HP home screen icon size for each control panel size. The following table lists the HP device control panel sizes and their respective display densities and recommended HP home screen icon sizes:

Physical Panel Width	Resolution (WxH px)	Display Density	HP Icon Size Category	Icon Size (WxH px)
10.1 inch 8 inch	1280x800 1024x768	mdpi	xl	179x179
8 inch	800x600	ldpi	lg	140x140
4.3 inch	480x272	ldpi	md	90x90

Include HP home screen icons as HP icon size category-specific drawables in your APK (e.g. res/drawable-lg/ic_hplauncher.png).

To help you, we have provided a comprehensive UI Style Guide.